Modul Praktikum Manajemen Basis Data

Konsep DBMS dan MySQL Dasar

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Tgl praktikum : 3 Desember 2024

Teknik Informatika

Universitas Muhammadiyah Bandung

2024

Mata Kuliah: Manajemen Basis Data
Durasi: 170 menit **Tujuan**:

- Mahasiswa memahami dan dapat menggunakan perintah dasar MySQL.
- Mahasiswa mampu menerapkan konsep database sederhana untuk memanipulasi data.

Materi dan Latihan

A. Pendahuluan

Pada praktikum ini, Anda akan membuat sebuah basis data sederhana untuk mengelola informasi mahasiswa. Praktikum mencakup:

1. Membuat database dan tabel.
2. Mengelola data menggunakan perintah dasar MySQL (CRUD).
3. Studi kasus sederhana.

B. Perintah Dasar MySQL

1. Membuat Database

Latihan :

- Buat database baru bernama universitas.

Perintah SQL:

```
create database universitas;
Use universitas;
```

Screenshot hasil running query

2. **Membuat Tabel**

Latihan:

- Buat tabel mahasiswa dengan kolom berikut: ○ nim (Nomor Induk Mahasiswa) sebagai Primary Key tipe varchar. ○ nama (Nama Mahasiswa) tipe varchar. ○ jurusan (Jurusan Mahasiswa) tipe varchar. ○ angkatan (Tahun Angkatan) tipe Year. ○ ipk (Indeks Prestasi Kumulatif) tipe decimal.

Perintah SQL:

```
create table mahasiswa(
        NIM varchar(10) primary
   key, nama varchar(50), jurusan
   varchar(50), angkatan year,
   ipk decimal
);
```

Screenshot hasil running query

3. **Menambahkan Data ke Tabel**

Latihan:

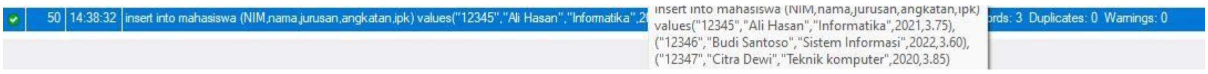
- Tambahkan 3 data mahasiswa berikut ke tabel mahasiswa.

Nim	Nama	jurusan	Angkatan	ipk
12345	Ali Hasan	Informatika	2021	3.75
12346	Budi Santoso	Sistem Informasi	2022	3.60
12347	Citra Dewi	Teknik Komputer	2020	3.85

Perintah SQL:

```
insert into mahasiswa (NIM,nama,jurusan,angkatan,ipk)
values("12345","Ali Hasan","Informatika",2021,3.75),
("12346","Budi Santoso","Sistem Informasi",2022,3.60),
("12347","Citra Dewi","Teknik komputer",2020,3.85);
```

Screenshot hasil running query



4. **Membaca Data dari Tabel**

Latihan:

- Tampilkan semua data mahasiswa.
- Tampilkan mahasiswa dari jurusan Informatika
- Tampilkan mahasiswa dengan IPK lebih dari 3.7 **Perintah SQL:**

- **Menampilkan semua data**

```
Select * from mahasiswa;
```

NIM	nama	jurusan	angkatan	ipk
12345	Ali Hasan	Informatika	2021	3.75
12346	Budi Santoso	Sistem Informasi	2022	3.6
12347	Citra Dewi	Teknik komputer	2020	3.85
NULL	NULL	NULL	NULL	NULL

- **Menampilkan mahasiswa dari jurusan "Informatika"**

```
Select * from mahasiswa where jurusan = "Informatika";
```

Screenshot hasil running query

NIM	nama	jurusan	angkatan	ipk
12345	Ali Hasan	Informatika	2021	3.75
NULL	NULL	NULL	NULL	NULL

- **Menampilkan mahasiswa dengan IPK lebih dari 3.7**

```
Select * from mahasiswa where ipk > 3.7;
```

Screenshot hasil running query

NIM	nama	jurusan	angkatan	ipk
12345	Ali Hasan	Informatika	2021	3.75
NULL	NULL	NULL	NULL	NULL

5. **Memperbarui Data**

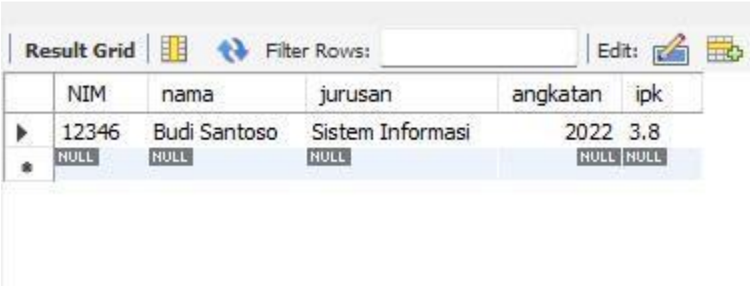
Latihan:

- Perbarui IPK mahasiswa dengan NIM '12346' menjadi 3.8.

Perintah SQL:

```
update mahasiswa set
ipk = 3.8 where NIM
= "12346";
```

Screenshot hasil running query



NIM	nama	jurusan	angkatan	ipk
12346	Budi Santoso	Sistem Informasi	2022	3.8
NULL	NULL	NULL	NULL	NULL

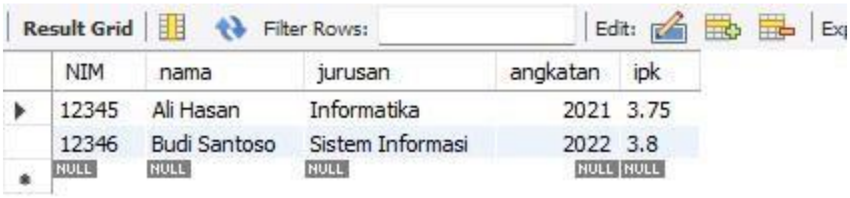
6. Menghapus Data

Latihan:

 Hapus data mahasiswa dengan Nama ‘Citra Dewi’. **Perintah SQL**

```
delete from mahasiswa where NIM = "12347";
```

Screenshot hasil running query



NIM	nama	jurusan	angkatan	ipk
12345	Ali Hasan	Informatika	2021	3.75
12346	Budi Santoso	Sistem Informasi	2022	3.8
NULL	NULL	NULL	NULL	NULL

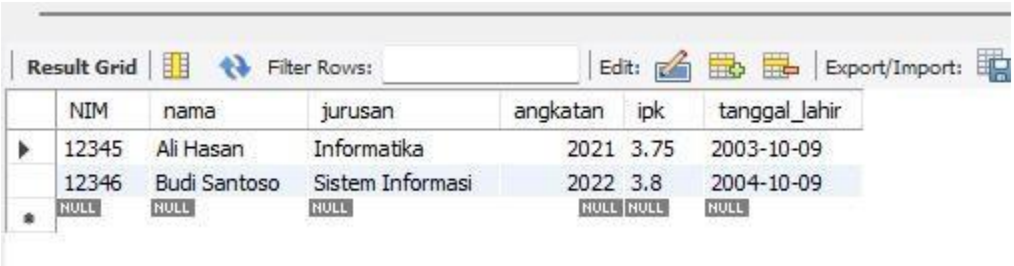
7. Mengubah struktur tabel Latihan

:

 Tambahkan kolom tanggal lahir pada tabel mahasiswa.
Masukkan nilai tanggal lahir pada setiap data di tabel mahasiswa
Perintah SQL :

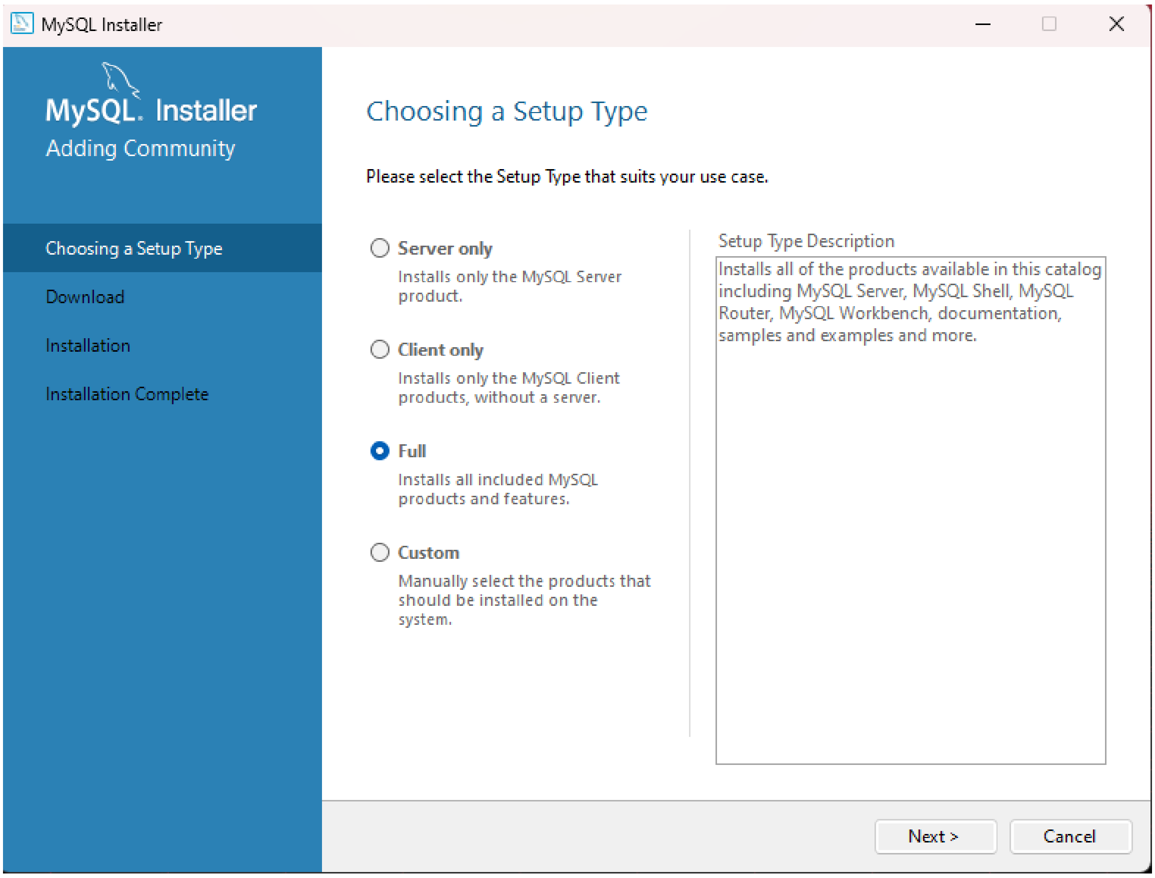
```
-         alter table mahasiswa add tanggal_lahir
date;
-         update mahasiswa set ipk = 3.8 where NIM
= "12346";
```

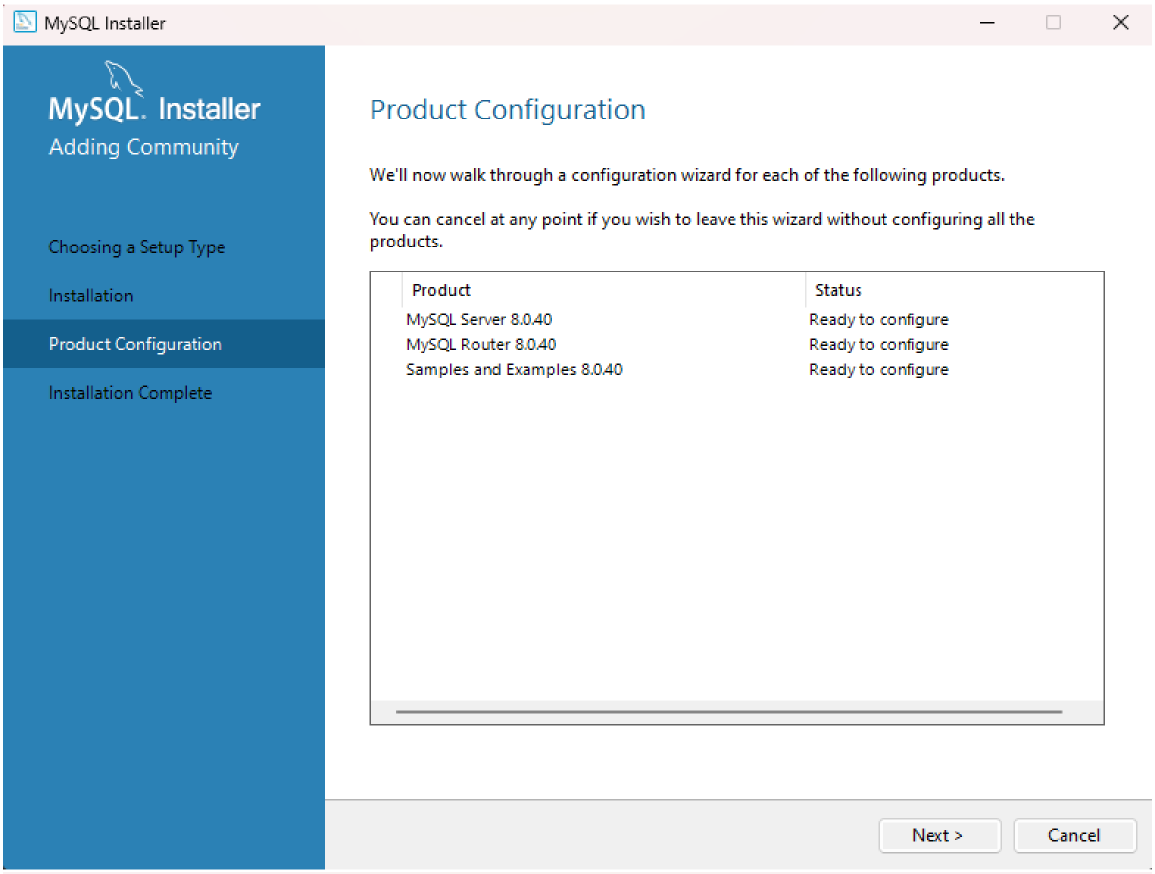
Screenshot hasil running query

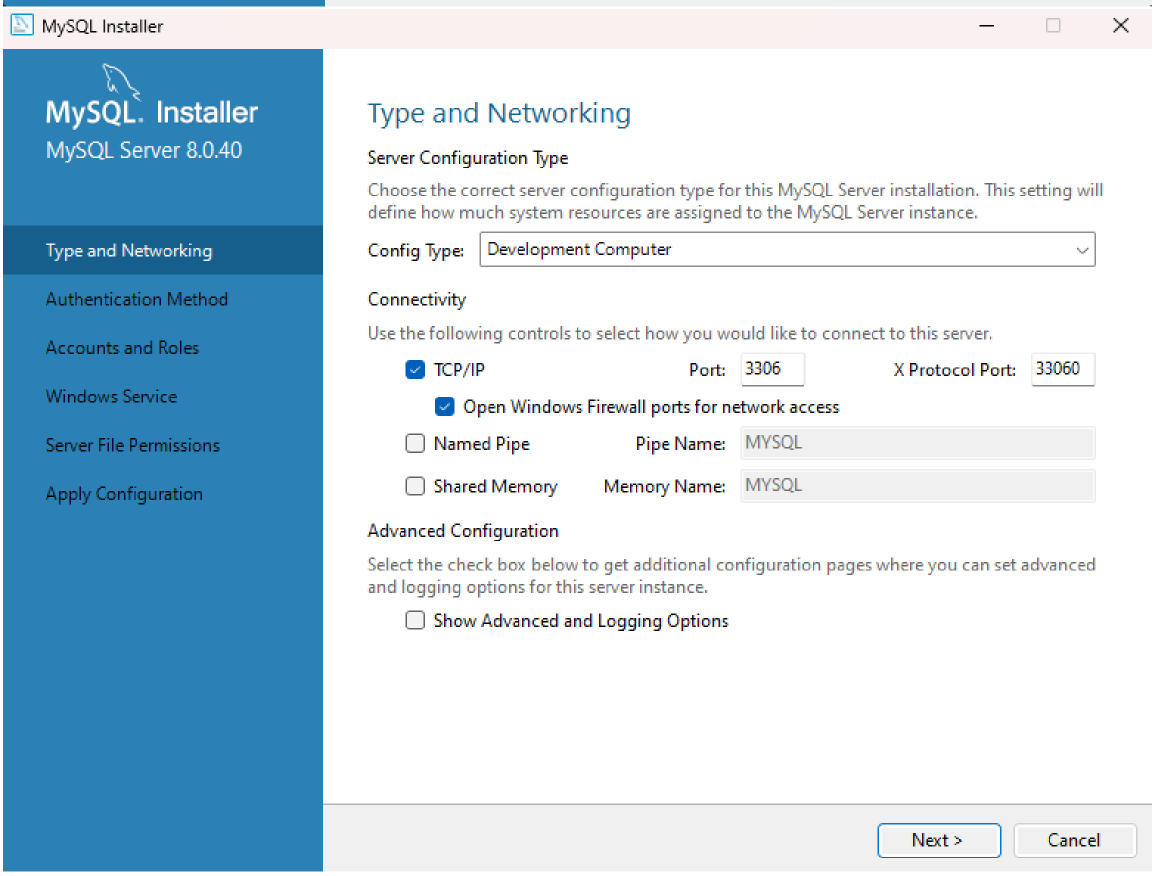


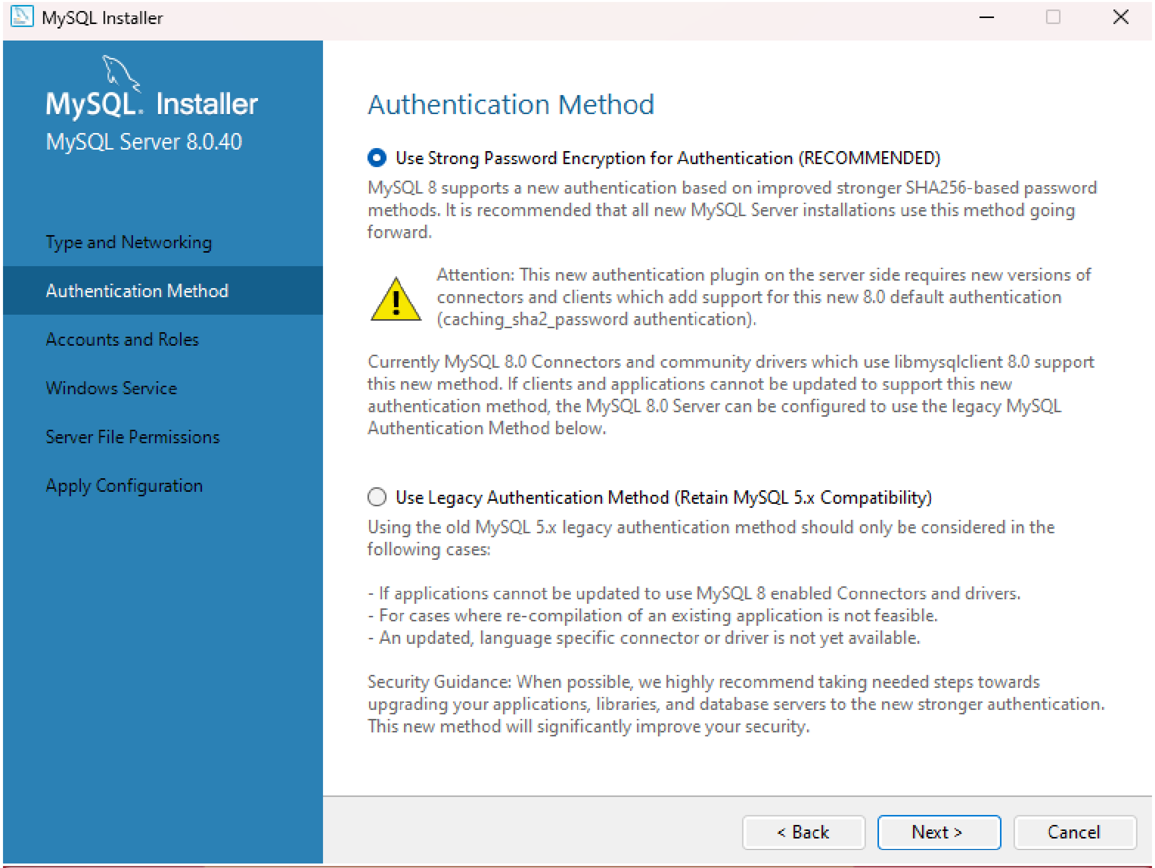
NIM	nama	jurusan	angkatan	ipk	tanggal_lahir
12345	Ali Hasan	Informatika	2021	3.75	2003-10-09
12346	Budi Santoso	Sistem Informasi	2022	3.8	2004-10-09
NULL	NULL	NULL	NULL	NULL	NULL

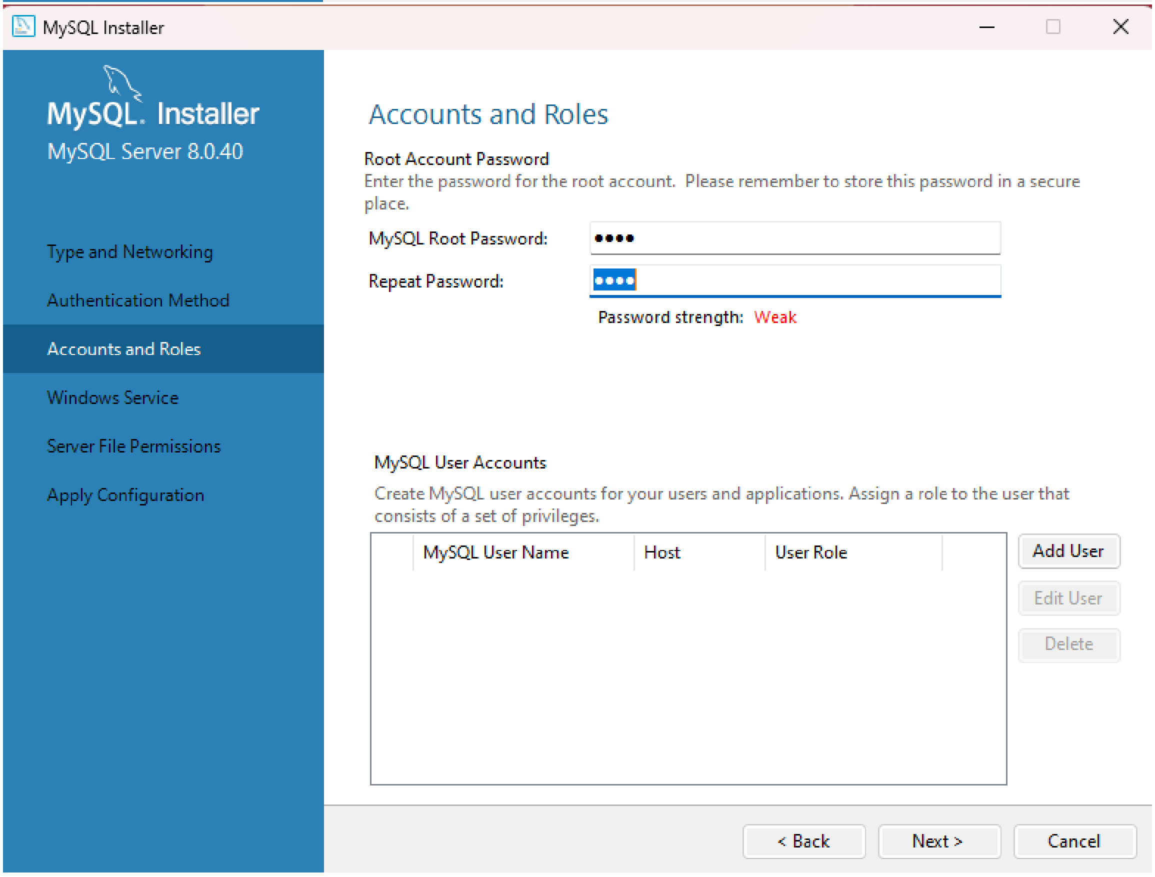
Dokumentasi Instalasi MySQL

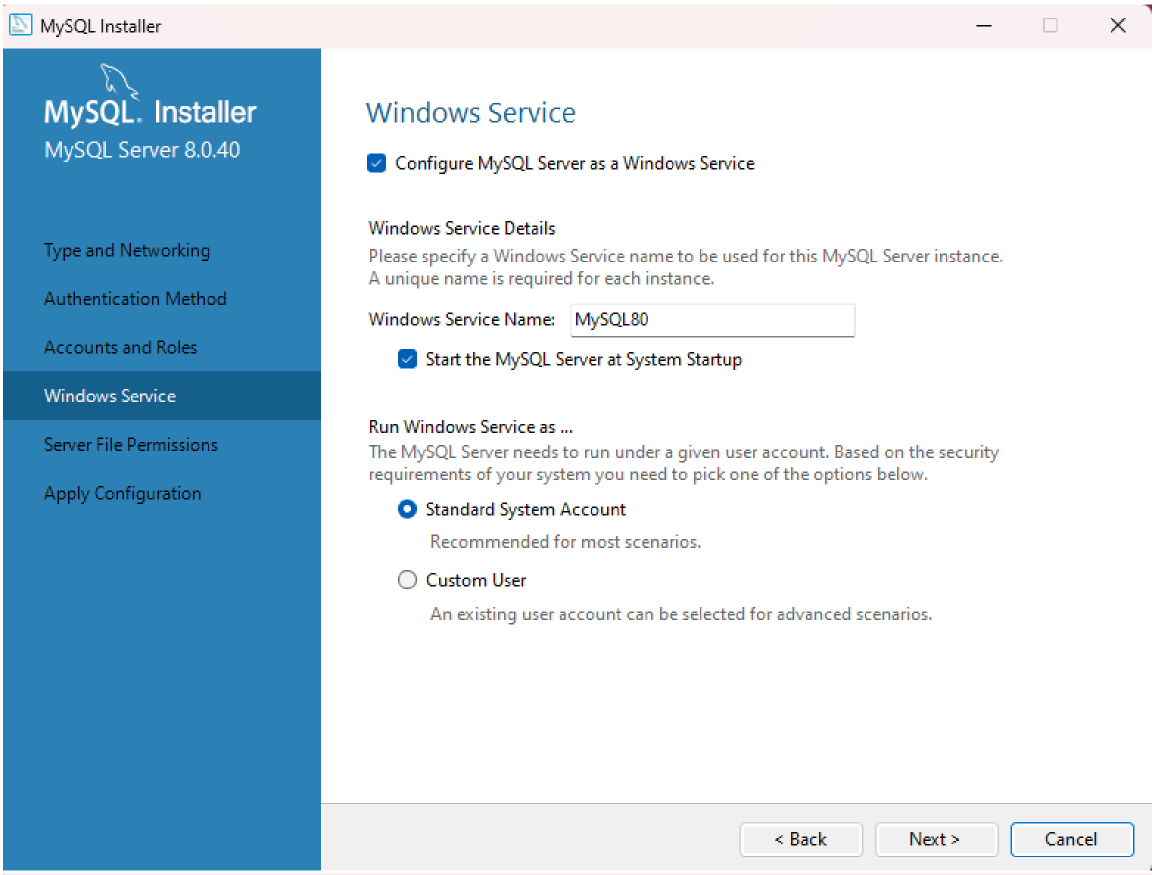


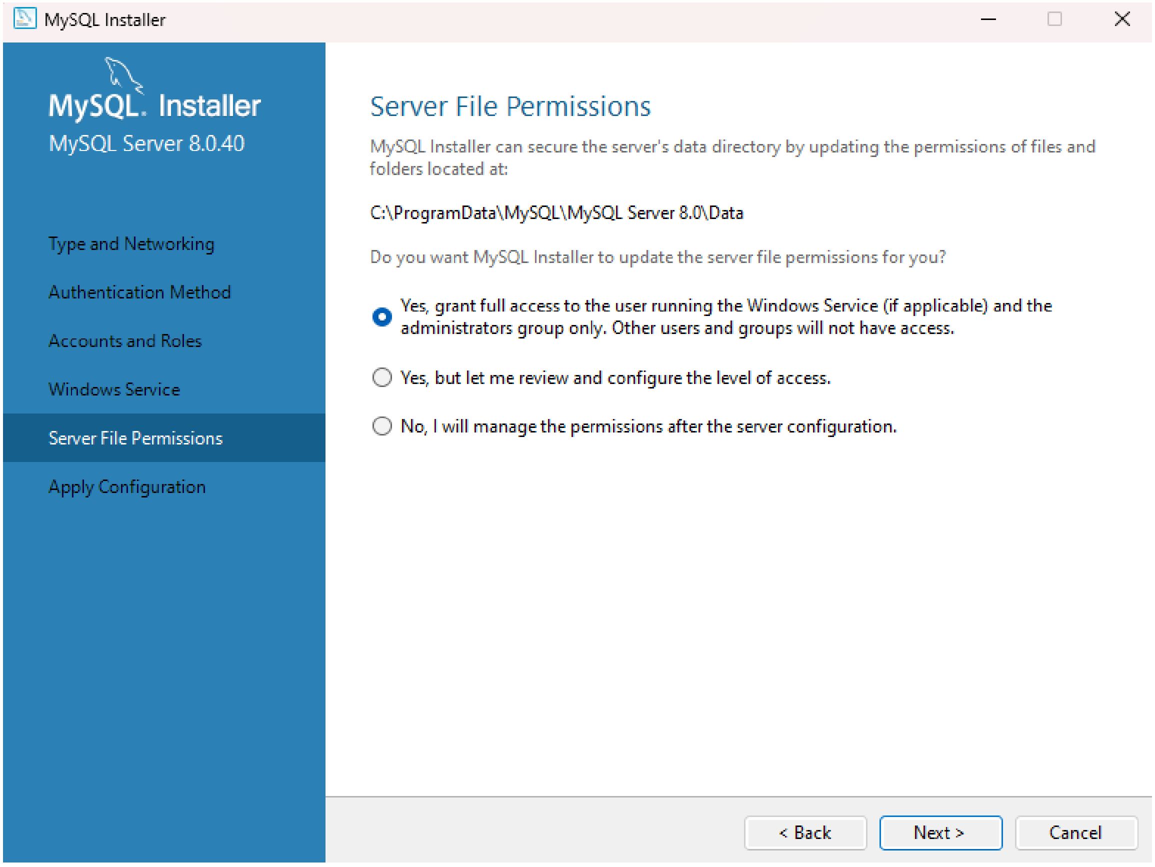

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.40
Ready to configure
MySQL Router 8.0.40
Ready to configure
Samples and Examples 8.0.40
Ready to configure
Next >
Cancel

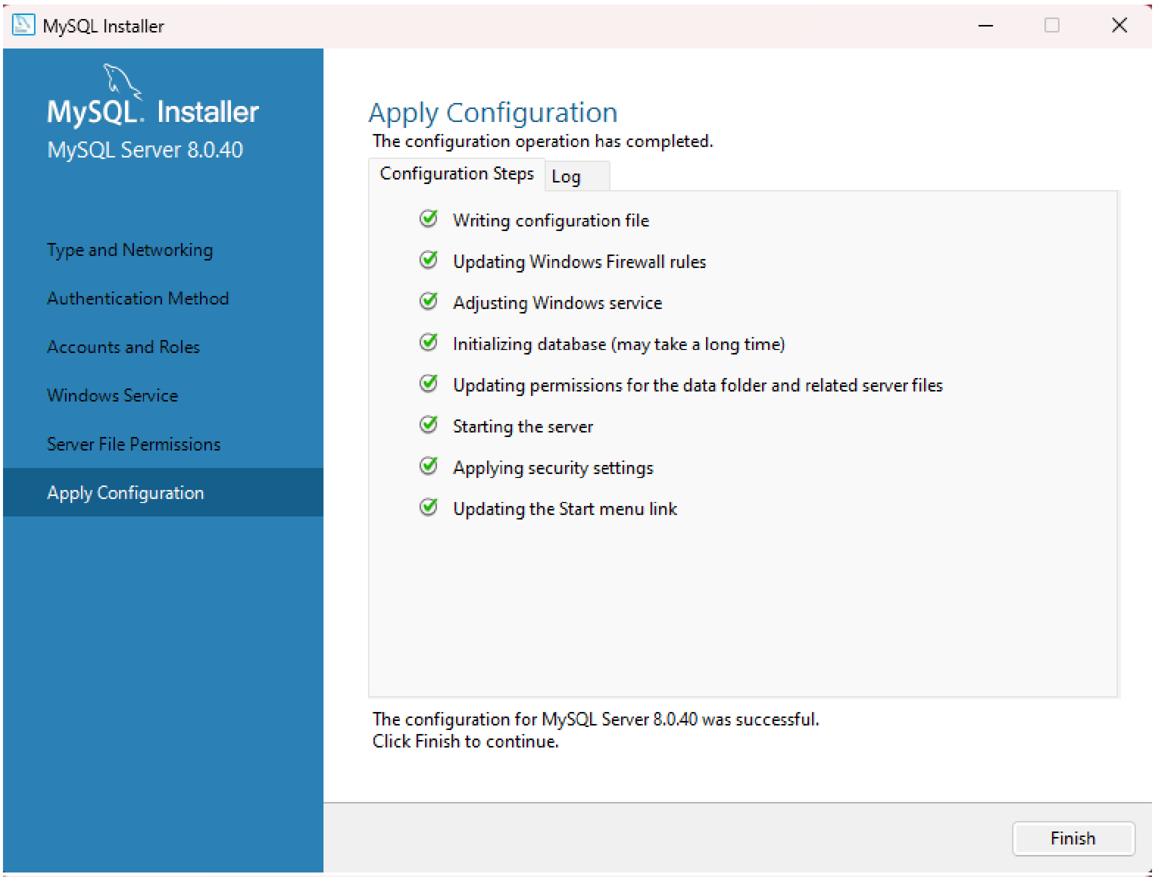

MySQL Installer
MySQL. Installer
MySQL Server 8.0.40
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Type and Networking
Server Configuration Type
Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.
Config Type:
Development Computer
Connectivity
Use the following controls to select how you would like to connect to this server.
TCP/IP
Port:
3306
X Protocol Port:
33060
Open Windows Firewall ports for network access
Named Pipe
Pipe Name:
MYSQL
Shared Memory
Memory Name:
MYSQL
Advanced Configuration
Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.
Show Advanced and Logging Options
Next >
Cancel

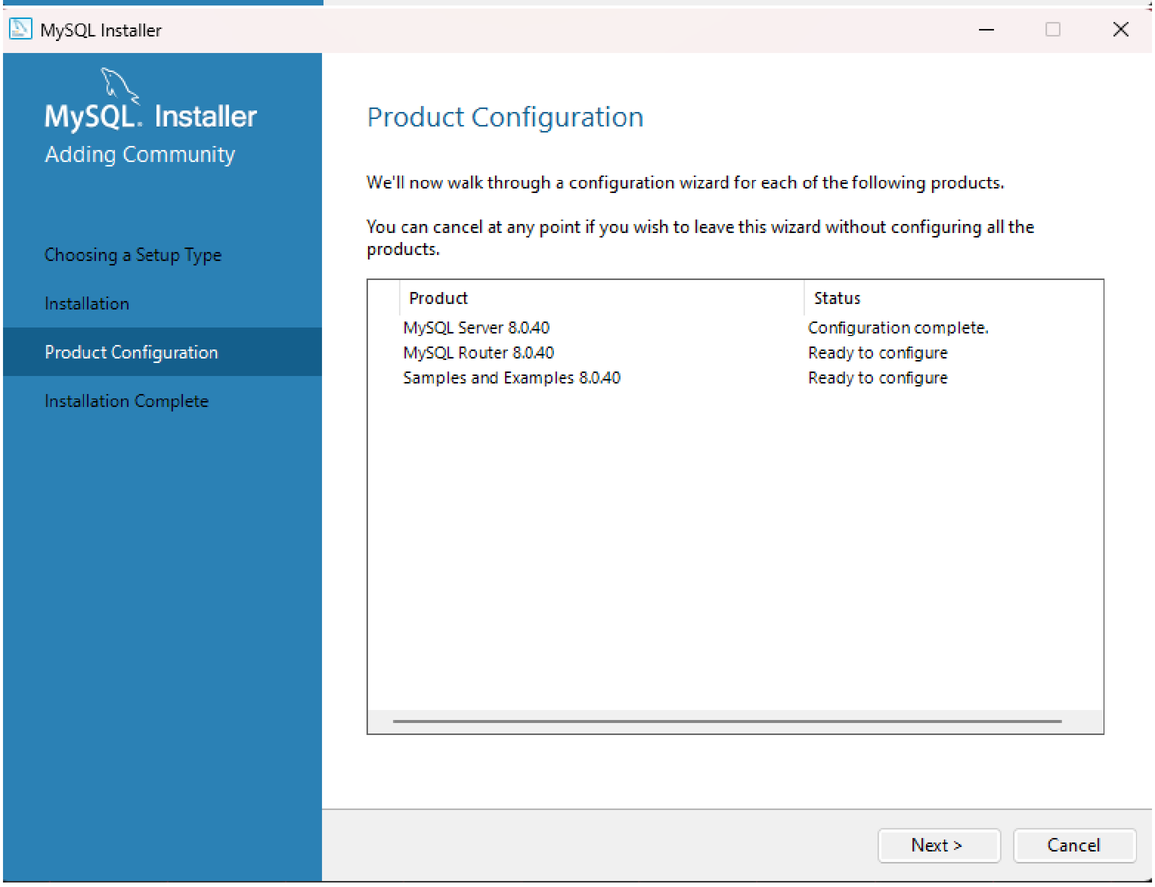

MySQL Installer
MySQL. Installer
MySQL Server 8.0.40
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Authentication Method
Use Strong Password Encryption for Authentication (RECOMMENDED)
MySQL 8 supports a new authentication based on improved stronger SHA256-based password methods. It is recommended that all new MySQL Server installations use this method going forward.
Attention: This new authentication plugin on the server side requires new versions of connectors and clients which add support for this new 8.0 default authentication (caching_sha2_password authentication).
Currently MySQL 8.0 Connectors and community drivers which use libmysqlclient 8.0 support this new method. If clients and applications cannot be updated to support this new authentication method, the MySQL 8.0 Server can be configured to use the legacy MySQL Authentication Method below.
Use Legacy Authentication Method (Retain MySQL 5.x Compatibility)
Using the old MySQL 5.x legacy authentication method should only be considered in the following cases:
- If applications cannot be updated to use MySQL 8 enabled Connectors and drivers.
- For cases where re-compilation of an existing application is not feasible.
- An updated, language specific connector or driver is not yet available.
Security Guidance: When possible, we highly recommend taking needed steps towards upgrading your applications, libraries, and database servers to the new stronger authentication. This new method will significantly improve your security.
< Back
Next >
Cancel

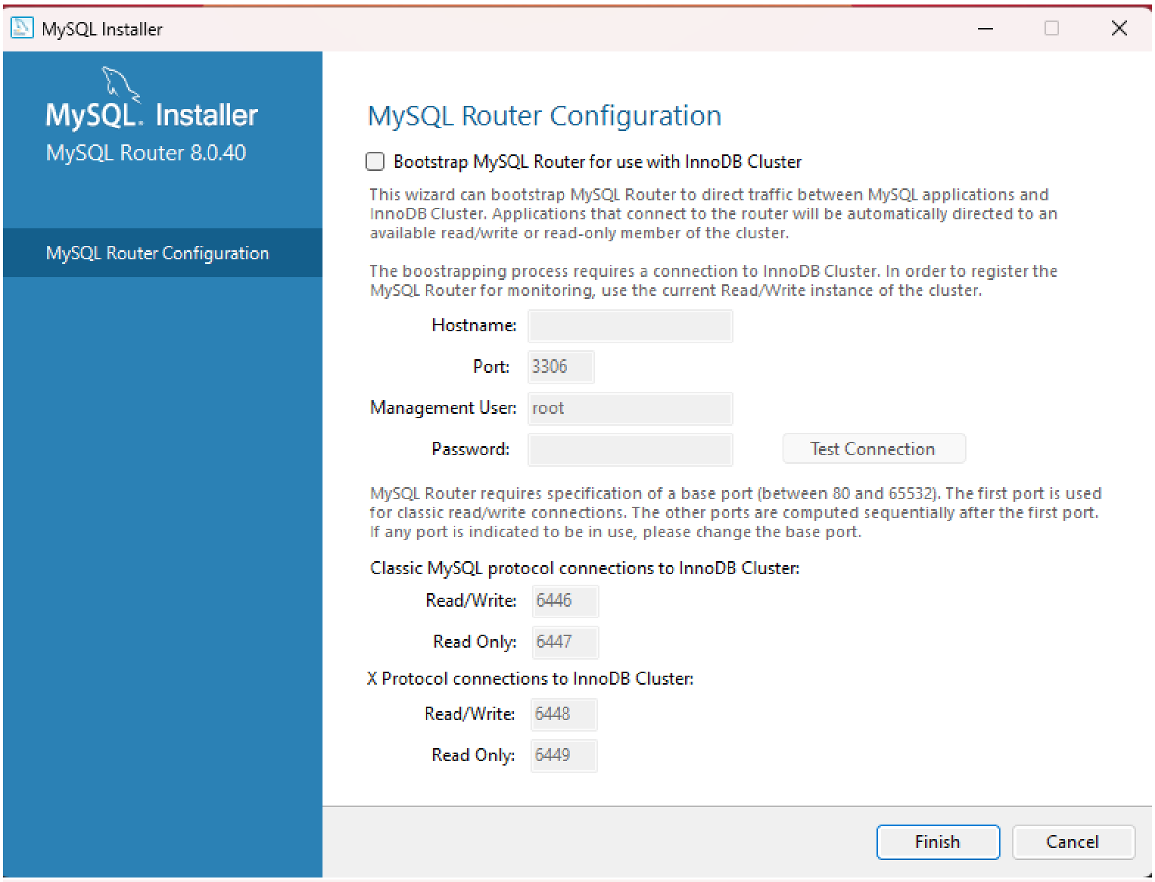

MySQL Installer
MySQL. Installer
MySQL Server 8.0.40
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Accounts and Roles
Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.
MySQL Root Password:
Repeat Password:
Password strength: Weak
MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.
MySQL User Name
Host
User Role
Add User
Edit User
Delete
< Back
Next >
Cancel

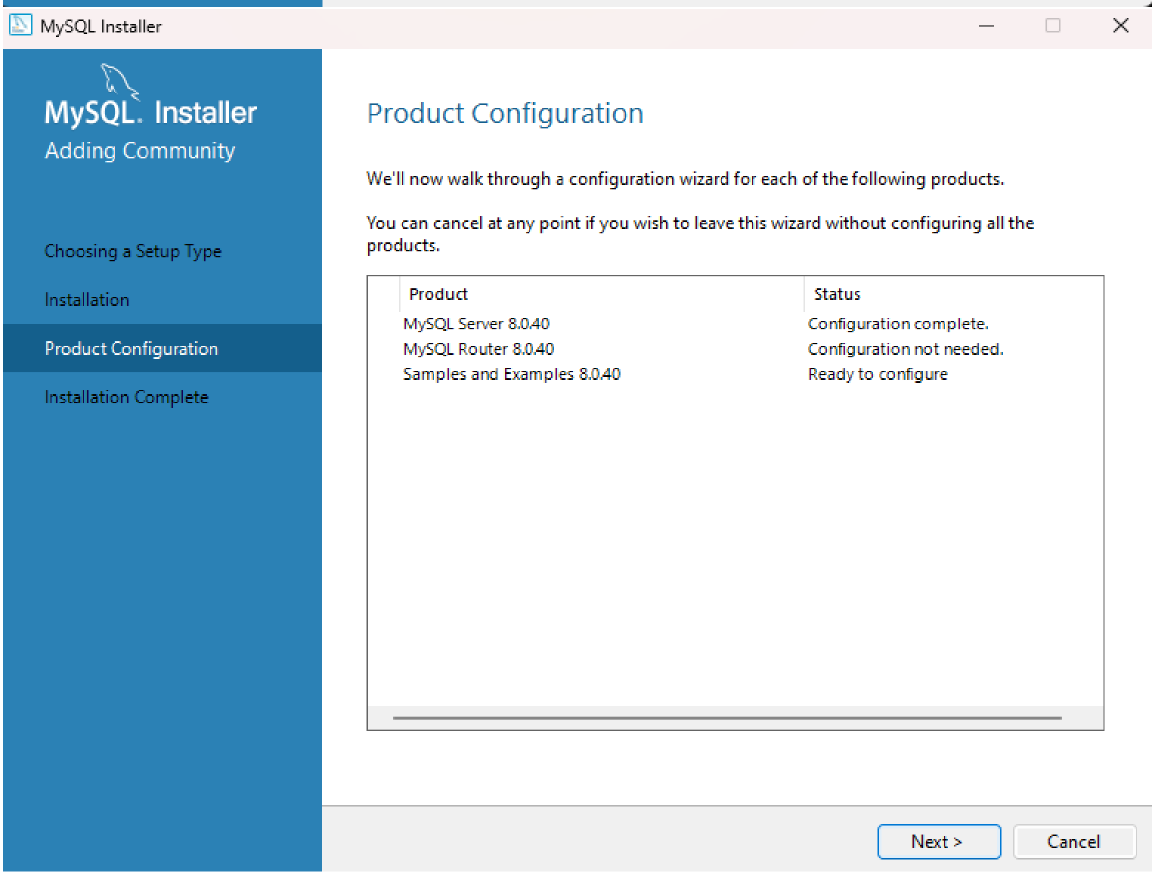

MySQL Installer
MySQL. Installer
MySQL Server 8.0.40
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Windows Service
Configure MySQL Server as a Windows Service
Windows Service Details
Please specify a Windows Service name to be used for this MySQL Server instance.
A unique name is required for each instance.
Windows Service Name: MySQL80
Start the MySQL Server at System Startup
Run Windows Service as ...
The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.
Standard System Account
Recommended for most scenarios.
Custom User
An existing user account can be selected for advanced scenarios.
< Back
Next >
Cancel

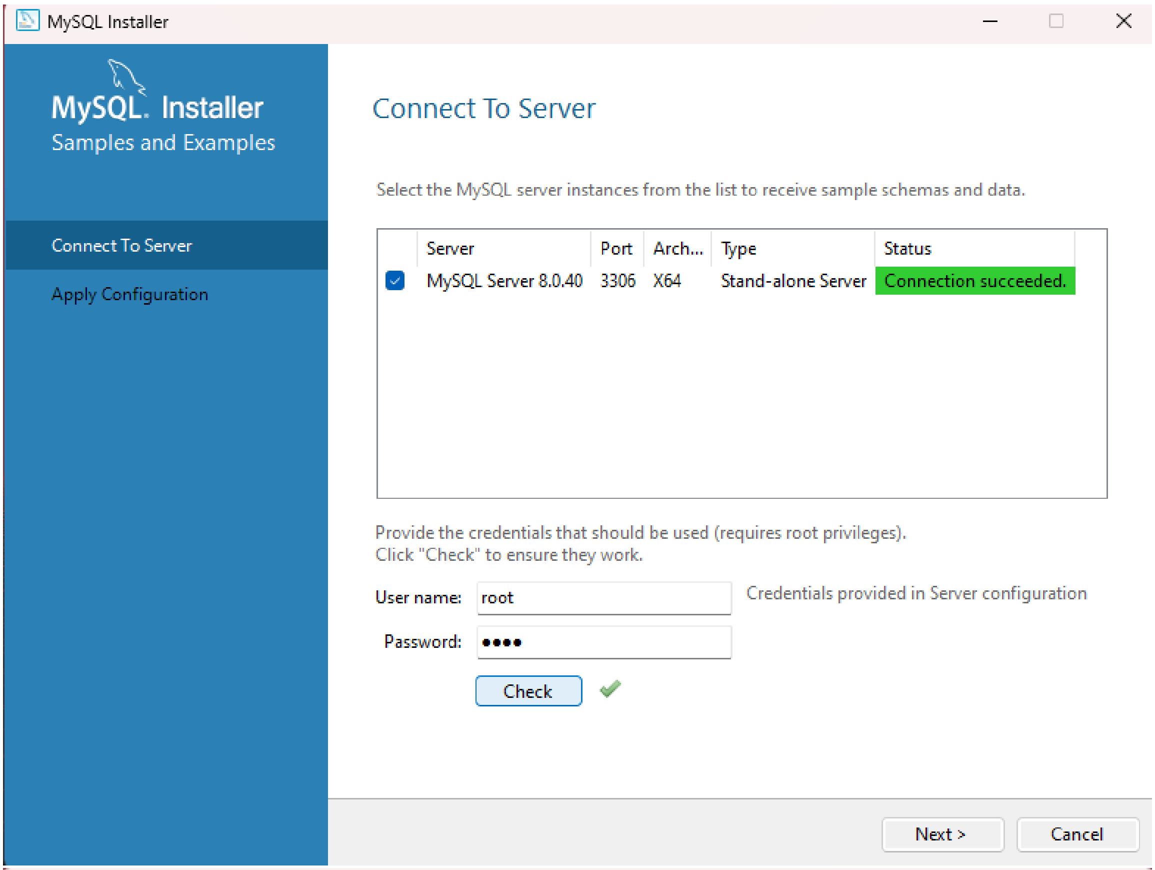

MySQL Installer
MySQL. Installer
MySQL Server 8.0.40
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Server File Permissions
MySQL Installer can secure the server's data directory by updating the permissions of files and folders located at:
C:\ProgramData\MySQL\MySQL Server 8.0\Data
Do you want MySQL Installer to update the server file permissions for you?
Yes, grant full access to the user running the Windows Service (if applicable) and the administrators group only. Other users and groups will not have access.
Yes, but let me review and configure the level of access.
No, I will manage the permissions after the server configuration.
< Back
Next >
Cancel

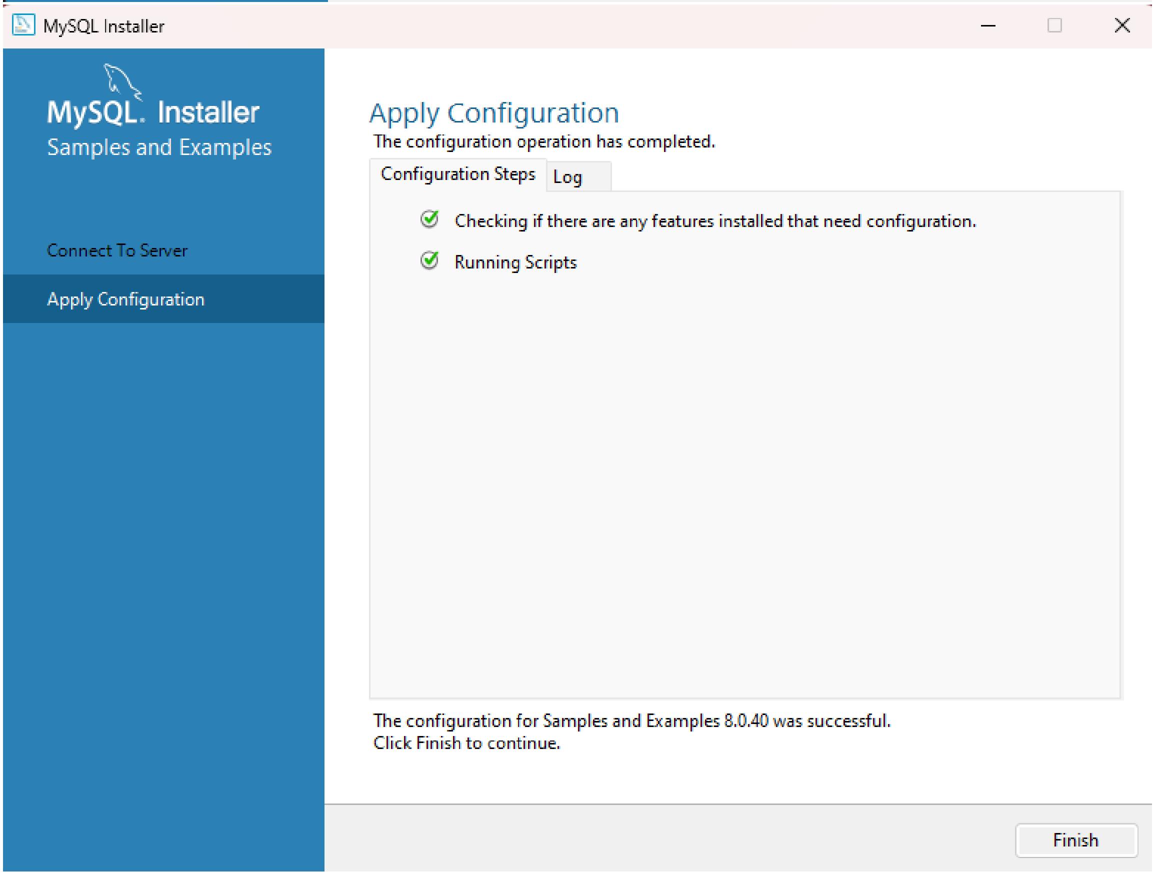

MySQL Installer
MySQL. Installer
MySQL Server 8.0.40
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
The configuration operation has completed.
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
The configuration for MySQL Server 8.0.40 was successful.
Click Finish to continue.
Finish

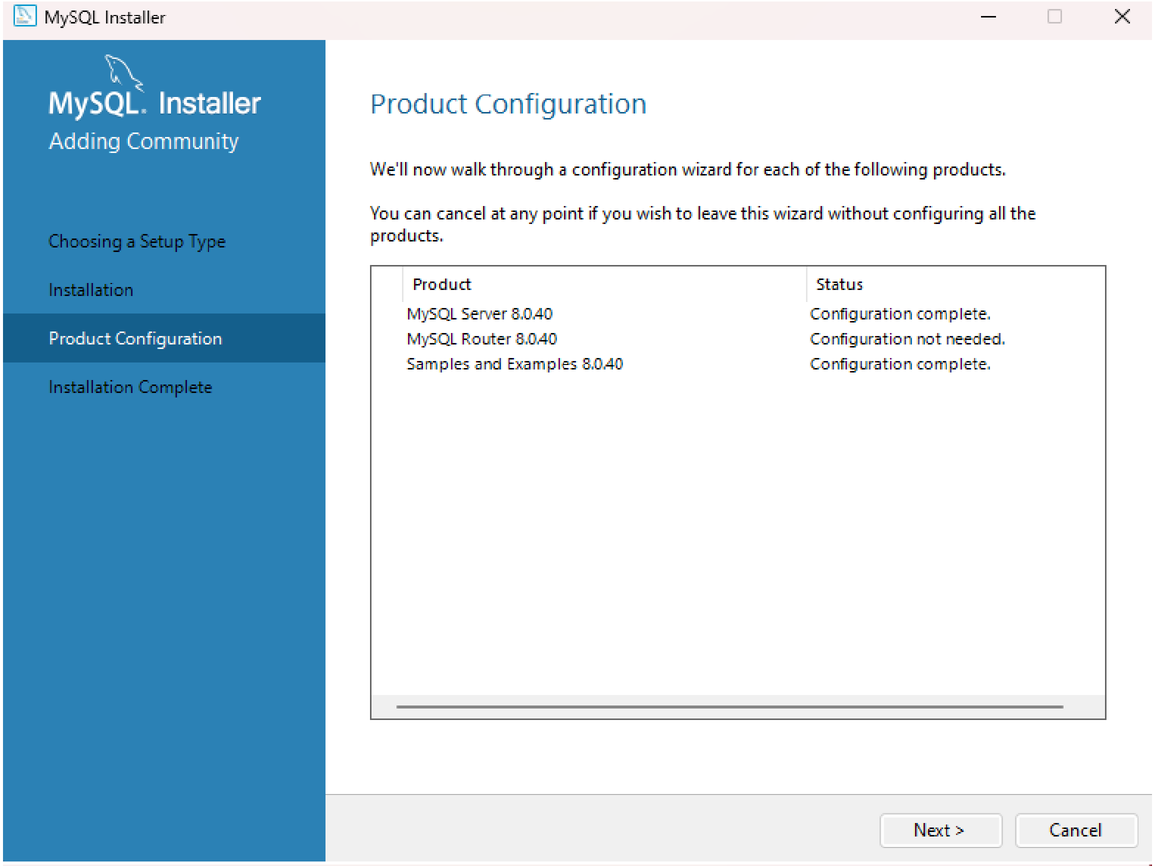

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.40
Configuration complete.
MySQL Router 8.0.40
Ready to configure
Samples and Examples 8.0.40
Ready to configure
Next >
Cancel

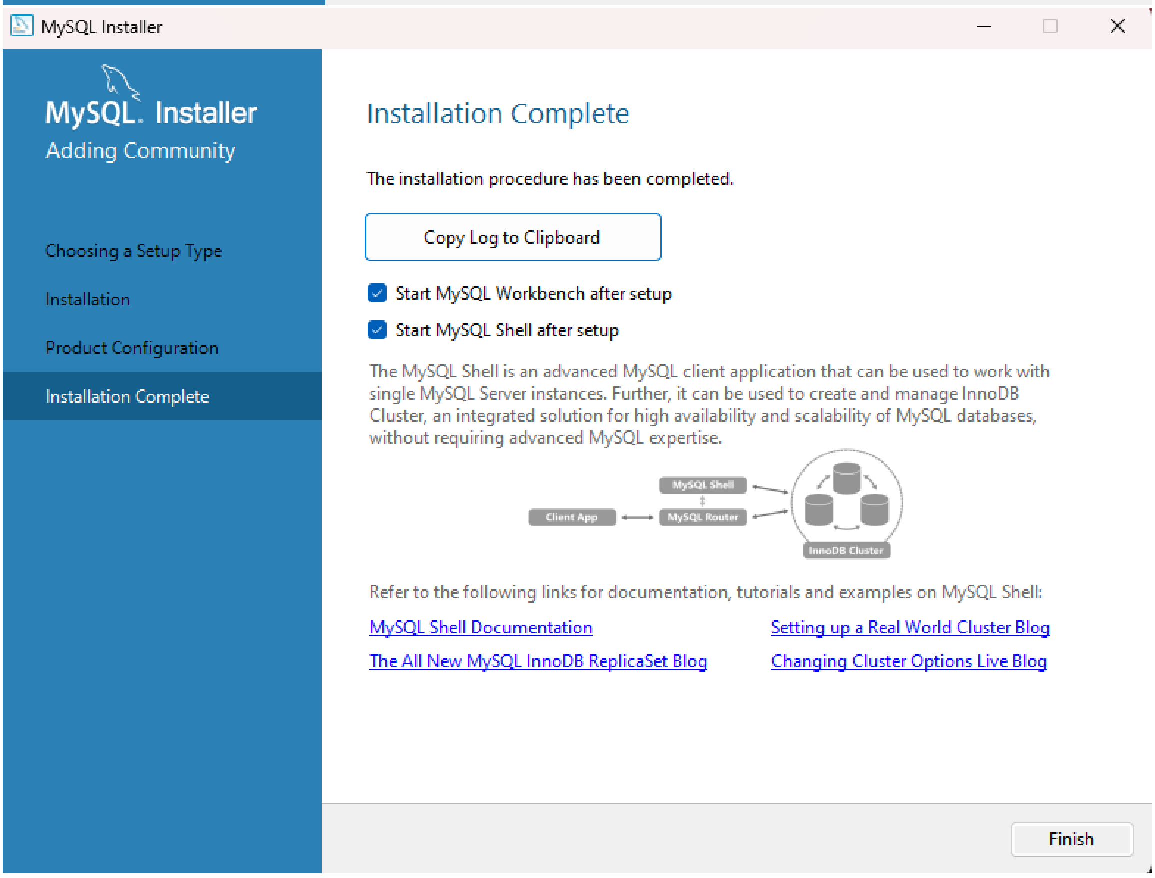

MySQL Installer
MySQL. Installer
MySQL Router 8.0.40
MySQL Router Configuration
MySQL Router Configuration
Bootstrap MySQL Router for use with InnoDB Cluster
This wizard can bootstrap MySQL Router to direct traffic between MySQL applications and InnoDB Cluster. Applications that connect to the router will be automatically directed to an available read/write or read-only member of the cluster.
The boostrapping process requires a connection to InnoDB Cluster. In order to register the MySQL Router for monitoring, use the current Read/Write instance of the cluster.
Hostname:
Port: 3306
Management User: root
Password:
Test Connection
MySQL Router requires specification of a base port (between 80 and 65532). The first port is used for classic read/write connections. The other ports are computed sequentially after the first port. If any port is indicated to be in use, please change the base port.
Classic MySQL protocol connections to InnoDB Cluster:
Read/Write: 6446
Read Only: 6447
X Protocol connections to InnoDB Cluster:
Read/Write: 6448
Read Only: 6449
Finish
Cancel


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.40
Configuration complete.
MySQL Router 8.0.40
Configuration not needed.
Samples and Examples 8.0.40
Ready to configure
Next >
Cancel


MySQL Installer
MySQL. Installer
Samples and Examples
Connect To Server
Apply Configuration
Connect To Server
Select the MySQL server instances from the list to receive sample schemas and data.
Server
Port
Arch...
Type
Status
MySQL Server 8.0.40
3306
X64
Stand-alone Server
Connection succeeded.
Provide the credentials that should be used (requires root privileges).
Click "Check" to ensure they work.
User name:
root
Credentials provided in Server configuration
Password:
Check
Next >
Cancel


MySQL Installer
MySQL. Installer
Samples and Examples
Connect To Server
Apply Configuration
Apply Configuration
The configuration operation has completed.
Configuration Steps
Log
Checking if there are any features installed that need configuration.
Running Scripts
The configuration for Samples and Examples 8.0.40 was successful.
Click Finish to continue.
Finish


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.40
Configuration complete.
MySQL Router 8.0.40
Configuration not needed.
Samples and Examples 8.0.40
Configuration complete.
Next >
Cancel


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Installation Complete
The installation procedure has been completed.
Copy Log to Clipboard
Start MySQL Workbench after setup
Start MySQL Shell after setup
The MySQL Shell is an advanced MySQL client application that can be used to work with single MySQL Server instances. Further, it can be used to create and manage InnoDB Cluster, an integrated solution for high availability and scalability of MySQL databases, without requiring advanced MySQL expertise.
MySQL Shell
Client App
MySQL Router
InnoDB Cluster
Refer to the following links for documentation, tutorials and examples on MySQL Shell:
MySQL Shell Documentation
Setting up a Real World Cluster Blog
The All New MySQL InnoDB ReplicaSet Blog
Changing Cluster Options Live Blog
Finish